Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/05/2018
PI Name:	Levine, Benjamin D M.D.		
Project Title:	STOP VIIP Study: Safe and Effective Countermeasures to Reduce Intracranial Pressure and Ameliorate VIIP in Astronauts		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	75231-5129	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2015	**End Date:**	05/31/2017
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lawley, Justin Ph.D. (The University of Texas Southwestern Medical Center) Whitworth, Louis M.D. (The University of Texas Southwestern Medical Center) Zhang, Rong Ph.D. (The University of Texas Southwestern Medical Center)		
Grant/Contract No.:	NCC 9-58-CA00005		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The visual impairment observed in many astronauts after long duration spaceflight aboard the International Space Station is considered the most important mission threatening medical problem identified in the last decade of the space program. Between 2012 and 2014, NASA and NSBRI (National Space Biomedical Research Institute) have spent $3-4 million/year investigating the clinical course and underlying mechanisms of this condition, termed the Visual Impairment Intracranial Pressure syndrome (VIIP) because of its theoretical association with elevated intracranial pressure (ICP). During the previous funding cycle, the investigators for this application performed the first invasive measures of intracranial pressure during acute (head down tilt (HDT) bedrest, and parabolic flight) and chronic (24 hours HDT bedrest) changes in gravitational gradients simulating spaceflight in healthy humans and have made the following novel observations: 1) there is a large range of ICP experienced normally during changes in posture in daily life; 2) these changes occur within seconds of postural transients from upright to supine and then are stable over time in that position; 3) with acute assumption of simulated microgravity (HDT bedrest), ICP goes up slightly from the supine position, but over the next 24 hrs, returns towards, and sometimes below the supine value because of spatial compensation within the brain; 4) during true microgravity (parabolic flight), the acute reduction in central venous pressure reduces cerebral outflow resistance and ICP decreases compared to the supine position, though remains above the upright position in 1G. Together, these results have led us to hypothesize that the VIIP syndrome is NOT due to a pathologically elevated ICP. However because there is not the usual reduction in ICP typically seen in the upright position on Earth, the incessant, low level elevation in ICP (i.e., greater than upright, but less than supine) leads to adaptive changes in the eye which ultimately result in globe flattening, choroidal folds, and visual impairment. Preliminary data suggest that the use of lower body negative pressure (LBNP) can lower ICP while in the head down position, towards the values observed upright. We propose that applying LBNP during sleep in space would be a practical strategy to restore a relevant (though reversed) circadian variability in ICP that would then prevent adaptive changes in the eye. The global objective of this project is to propose, test, and validate this pathophysiologically based countermeasure that holds promise to eliminate the VIIP syndrome. We propose to test the following hypotheses: Hypothesis 1: LBNP will reduce supine ICP towards that observed in the upright posture. To test this hypothesis, we will accomplish specific aim #1: Specific Aim 1: Using direct invasive recording of ICP, we will determine the optimal level of LBNP required to cause a sustained reduction in ICP towards that observed in the upright position, while avoiding hypotension and maintaining cerebral perfusion. Hypothesis 2: Repeated administration of LBNP at night will cause a safe, sustained reduction in ICP while asleep. This restoration of effective circadian variability in ICP will prevent structural changes in the eye induced by microgravity. To test this hypothesis, we will accomplish Specific Aim #2: Specific Aim 2: We will examine the long-term efficacy of nightly LBNP (6-8 hrs) to simulate the upright position and ICP circadian variability and prevent structural changes in the eye during multiple (4) days of HDT bedrest. We will use comprehensive, noninvasive state-of-the-art imaging to examine the structure of the visual apparatus, the intra- and extra-cranial veins, cerebral spinal fluid production, and localized intracranial fluid shifts using high-resolution physiological, optical, and magnetic resonance techniques, combined with invasive direct measurement of ICP.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In the setting of severe traumatic brain injury, elevations in ICP are common with sustained elevations in ICP predicting morbidity and mortality. Given the result presented in Aim #1, a simple candidate solution to lower ICP may be lower body negative pressure. We have recently received grant funding to examine the hypothesis that LBNP will reduce mean intracranial pressure, intracranial pulse amplitudes (surrogate of compliance), and the occurrence of intracranial pressure waves (B- & A-waves). As a consequence of improved intracranial stability, ICU discharge is hypothesized to be expedited with patients reporting lower degrees of disability at discharge and 6-month post discharge from hospital. It is possible that acute or prolonged periods of LBNP may be beneficial in other diseases of intracranial or cerebral venous hypertension. These areas of research will be the focus of our research effort over the coming years and are directly linked to our research funded by NSBRI.
Task Progress:	FINAL REPORTING IN AUGUST 2017 Specific aim #1. Specific aim #1 is complete. The data have been analyzed and are currently under final revision for peer review submission. Specific aim #2. Specific aim #2 is complete. The data are currently being analyzed and prepared for publication. ANNUAL REPORTING IN NOVEMBER 2016 Specific aim #1. Specific aim #1 is complete. The data has been analyzed and is being prepared for publication. Specific aim #2. We have obtained Institutional Review Board approval for this aim. Approval has been obtained and orders written for our subjects to be admitted to the CTRC (Clinical and Translational Research Center) at UT Southwestern for the bedrest portion of their studies and at the Advanced Imaging Research Center for magnetic resonance imagining. Due to the complex and invasive nature of this protocol, we have, and continue to undertake extensive pilot testing. A lightweight lower body negative pressure sleep sac is currently in development by our collaborators at Under Amour. We have worked through a number of prototypes and have confidence that a final version will be available in the near future. Nonetheless, to allow us to proceed with aim #2 in a timely fashion, we have sourced an alternative magnetic friendly lower body negative pressure chamber. We expect to finalize pilot testing of magnetic resonance imaging sequences including with and without LBNP in the next 4-6 weeks. Thereafter, we will be ready to begin testing research subjects. We have contacted a number of Ommaya patients who completed previous studies; they are engaged and looking forward to helping develop a countermeasure for visual impairment in astronauts.
Bibliography Type:	Description: (Last Updated: 03/19/2026)
Articles in Peer-reviewed Journals	Lawley JS, Petersen LG, Howden EJ, Sarma S, Cornwell WK, Zhang R, Whitworth LA, Williams MA, Levine BD. "Effect of gravity and microgravity on intracranial pressure." J Physiol. 2017 Mar 15;595(6):2115-27. Epub 2017 Feb 14. https://doi.org/10.1113/JP273557 ; PubMed PMID: 28092926; PubMed Central PMCID: PMC5350445 , Mar-2017

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Awards	Levine BD. "Best Doctors 2016 Cardiology, March 2016." Mar-2016
Awards	Levine BD. "Robert M. Berne Distinguished Lectureship, April 2017." Apr-2017